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Predicting Imminent Conversion to Exudative Age-Related Macular Degeneration Using Multimodal Data and Ensemble Machine Learning

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Objective: Exudative age-related macular degeneration (eAMD) is a major cause of central vision loss. Identifying patients at high risk of imminent eAMD could enable timely treatment and improve outcomes. Our goal was to develop and compare classical machine learning (ML) and deep learning (DL) models to predict imminent eAMD conversion within 6 months and integrate OCT with clinical data into a single predictive model.

Design: Retrospective cohort study.

Participants: Patients seen at the Wilmer Eye Institute between 2013 and 2021 with eAMD in ≥1 eye.

Methods: Spectral domain OCT volumes prior to conversion and the corresponding clinical data (age, best-corrected visual acuity, sex, and fellow-eye status) were collected and used for model training or testing. ResNet-50 and classical ML (Random Forest and XGBoost) models were trained to predict imminent conversion to eAMD within 6 months on an eye level. For the multilayer perceptron (MLP) framework, the trained ResNet-50 model was used as the feature encoder, and a downsampled feature vector concatenated with corresponding clinical tabular data was passed through the MLP (prediction head). Data were partitioned at the patient level (75% training, 15% validation, and 10% testing). Model performance was evaluated using the area under the operating characteristic curve (AUC) and 95% confidence interval (CI) for the model AUC was calculated using the percentile method after bootstrapping the test set 10 000 times. Model comparisons were made using modified paired *t* test. $P < 0.05$ was considered statistically significant.

Main Outcome Measures: Area under the operating characteristic curve.

Results: Thirty-three thousand one hundred eighty-nine OCT volumes from 2084 patients (63% female; 89.1% White, 4.8% Black, and 2.3% Asian) were included. The mean age at the time of first-eye conversion was 78.9 (± 9.3) years. Our best-performing models, "MLP multimodal" (trained with both OCT and clinical data; AUC: 0.76, 95% CI: 0.71–0.80) and "CNN OCT" (trained with only OCT data; AUC: 0.75, 95% CI: 0.70–0.79), had a DL (ResNet-50) architecture; "MLP multimodal" outperformed "CNN OCT" in predicting both all-eye ($P < 0.05$) and first-eye conversion ($P < 0.001$).

Conclusions: The 3-dimensional DL models, trained with OCT volumes, are capable of predicting both first-eye and fellow-eye imminent conversion to eAMD. The addition of clinical data further improved the model performance. These models, if validated prospectively, could serve as screening tools and allow retinal specialists to prioritize patients with more acute retinal issues.

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Age-related macular degeneration (AMD) is the leading cause of central vision loss in people older than 50 years in the United States.[1] It is estimated that 300 million people will have the disease by 2040 globally.[2] Approximately 90% of vision loss in AMD is due to exudative AMD (eAMD). Being able to identify patients at a high risk for imminent conversion to eAMD is crucial, as better presenting visual acuity (VA) at the time of treatment initiation with anti-VEGF injections predicts better final VA.[3] Currently, the average risk over 5 years of progression to advanced AMD, including eAMD, can be estimated using Age-Related Eye Disease Study criteria, such as drusen size and presence of pigmentary abnormalities.[4,5] The advent of OCT imaging has further broadened our armamentarium in predicting AMD progression. For example, various OCT biomarkers, including increased central drusen volume,[6,7] intraretinal hyperreflective foci,[8,9] drusen with hyporeflective core,[10,11] and subretinal drusenoid

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deposits[12,13] are now known to be risk factors for AMD progression.[14]

Although the Age-Related Eye Disease Study criteria allow us to broadly risk-stratify patients with AMD, they cannot provide fine-grained predictions for which patients are at imminent risk of conversion to eAMD. A promising approach to address this limitation is to leverage the micron-level resolution of OCT images and the pattern recognition of deep learning (DL), the cutting-edge machine learning (ML) technique, for medical image analysis. In fact, in a landmark paper, Yim et al demonstrated that imminent conversion to eAMD on an eye level can be predicted by a DL model trained with OCT images. Specifically, imminent conversion to eAMD within 6 months in the fellow eye of patients with known eAMD in the first eye could be predicted with an area under the operating characteristic curve (AUC) of 0.745 to 0.886.[15]

Building on the existing literature, we developed and compared different ways to combine OCT images with clinical variables into a single model, using both classical ML and DL architectures, to predict imminent (within 6 months) conversion to eAMD for both first- and fellow-eye conversions. We believe that the ability in predicting first-eye conversions is particularly important from a public health point of view, as this could allow monitoring of patients with early and intermediate AMD at an optometry or even primary care setting, decompress the patient back-log at ophthalmology practices, and allow ophthalmologists and retinal specialists to prioritize patients with more acute ophthalmic issues.

Methods

Patient Data

This is a retrospective study that adheres to the tenets of the Declaration of Helsinki and was approved by the Johns Hopkins University Institutional Review Board. Because the study was retrospective, informed consent was not required. Records of patients, seen at the Wilmer Eye Institute Johns Hopkins Medicine between 2013 and 2021, were reviewed. The inclusion criteria were conversion to eAMD in $\geq$1 eye and availability of spectral domain OCT images (Spectralis) obtained prior to conversion. The exclusion criteria were concurrent significant macular pathologies that were not AMD-related, unavailability of OCT images prior to conversion, and OCT images that were not centered on the macula.

For eyes that converted to eAMD, only OCT images obtained before conversion were included, and these OCT images were labeled as either "more than 6 months from conversion" or "within 6 months of conversion." We defined the conversion date as the "date of the first anti-VEGF injection." For eyes that never converted to eAMD, only OCT scans obtained more than 6 months prior to the last available OCT scan were included, and these OCT scans were labeled as "more than 6 months from conversion." The number of line scans within each OCT volume ranged from 25 to 97. The majority of OCT volumes (> 85%) contained 49 line scans. In addition to the OCT scans, the following clinical variables corresponding to each individual OCT scan were collected: age at the time of scan, best-corrected visual acuity (BCVA) at the time of scan, sex, and fellow-eye status (yes vs. no eAMD).

Data Split for Model Training

The models were trained with the following 3 different approaches:(1) with 3-dimensional (3D) OCT volumes alone (Fig 1A); (2) with clinical data alone (Fig 1B); and (3) with both 3D OCT volumes and clinical data (Fig 1C, D). For each model, the data split at the patient level was 75% training, 15% validation, and 10% testing. All models used the same training, validation, and testing splits.

Approach 1: Training a Convolutional Neural Network with Only OCT Volumes. We compared the performance of 7 3D convolutional neural network (CNN) models: ResNet-18, ResNet-34, ResNet-50, ResNet-101, MobileNet, GoogleNet, and DenseNet. Using transfer learning, these models were fine-tuned with the ImageNet weights as starting points. During fivefold cross-validation (CV) and using only data in the training and validation splits, we compared the mean and standard deviation of the AUC across the fivefolds for each model and identified the CNN backbone with the most robust performance—ResNet-50.[16] The ResNet-50 model was used for all the following subsequent experiments: balanced cross entropy loss (gamma = 0, alpha = 0.8), Adam optimizer, learning rate of 0.001, momentum of 0.9, and 100 epochs.

Approach 2: Training Classical Machine Learning Models Using Clinical Data Only. Two traditional ML classifiers, Random Forest (RF) and XGBoost (Fig 1B), were trained using the following clinical variables only: BCVA, sex, age, and fellow-eye status. Synthetic minority oversampling technique was used to address class imbalance eyes within 6 months of conversion and eyes more than 6 months away from conversion.

The following RF model hyperparameters were tuned using the scikit-learn package Randomized Search CV method: maximum depth, the maximum of features tested by 1 tree, the minimum number of samples needed to start a new leaf node, the minimum number of samples to split a node, the number of trees to make, and if bootstrapping should be used. A randomized search with 200 iterations was performed over the grid of these parameters with threefold cross-validation. Further tuning was performed by creating ranges around the best random fit parameters found and performing a full grid search of the resulting grid of parameters. Hyperparameter tuning for the XGBoost model was performed with a grid search of the possible values (including the default) and a threefold CV. The parameters tested were as follows: the learning rate, the minimum loss reduction to split a node, the maximum depth, the minimum hessian weight needed for a child tree, and the ratio of subsamples in the training set.

Approach 3: Training with Models with Both 3D OCT Volumes and Clinical Data. Two model frameworks were trained and tested, the classical ML models (Fig 1C) and a multilayer perceptron (MLP) neural network (Fig 1D).

For the classical ML models (RF and XGBoost), the 3D OCT volume classification features were extracted from the final fully connected layer (2048 features) of the ResNet-50. The extracted image feature vector and the 4 clinical variables (BCVA, sex, age, and fellow-eye status) were then concatenated into a single feature vector. Positive cases (conversion within 6 months) were upsampled using the synthetic minority oversampling technique. For both the RF and XGBoost models, default hyperparameters were used.

For the MLP framework, the trained 3D ResNet-50 CNN was used as the feature encoder and the MLP was used as the prediction head (Fig 1D).

The features from the final fully connected layer were downsampled from 2048 features to 10 features. This downsampled feature vector was concatenated with corresponding clinical tabular data and passed through the MLP. The MLP consisted of 3 linear transformation layers, a batch normalization layer, and a rectified

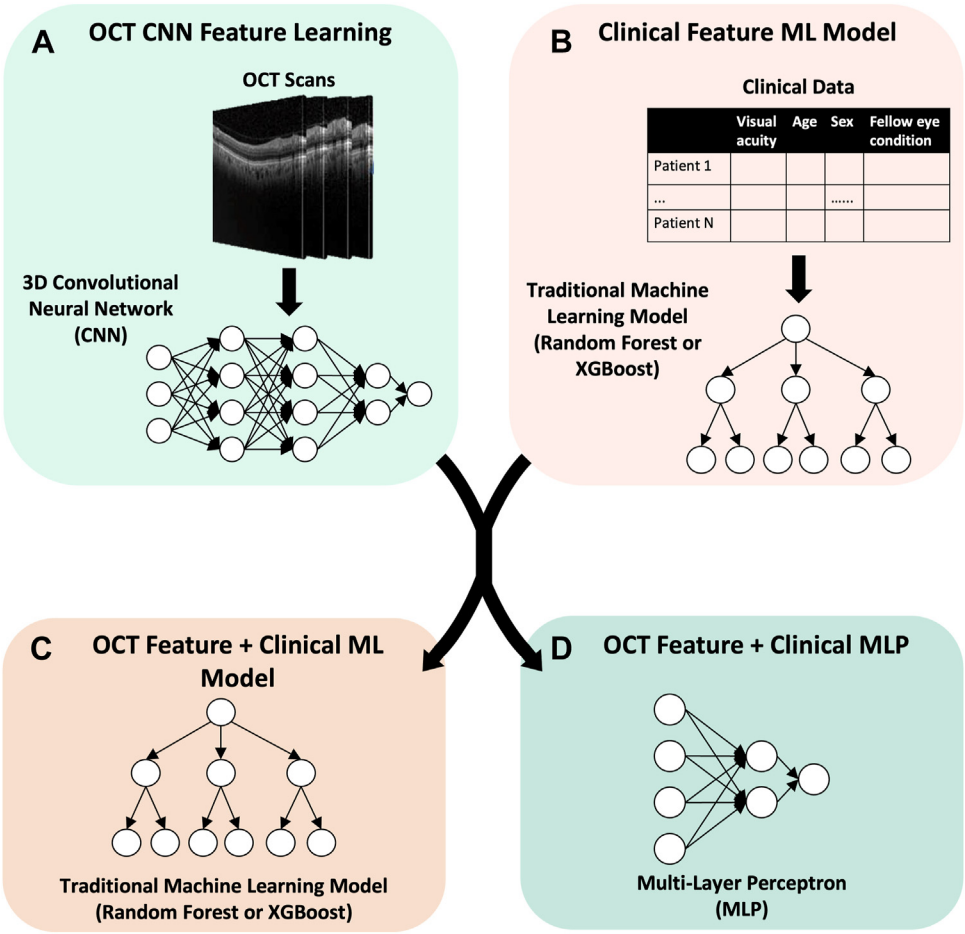


Figure 1. Overview of model development. **A,** A 3-dimensional convolutional neural network was trained using OCT volumes. **B,** Classical ML models (Random Forest and XGBoost) were trained using clinical data. **C,** The 2048-dimension feature vector from the CNN was included as predictors with clinical data in a Random Forest or XGBoost model. **D,** Alternatively, the clinical data were concatenated to the CNN feature vector and passed through a multilayer perceptron. CNN = convolutional neural network; ML = machine learning.

linear unit nonlinear activation layer to reduce the dimensionality of the feature map to perform binary classification. The pretrained 3D ResNet-50 feature encoder was frozen during training of the MLP to retain the learned feature images and focus the training on the combination of the image features and clinical tabular data. The final model was trained with the following hyperparameters: Adam optimizer, a learning rate of 1×10^{-6}, balanced cross entropy loss, and 500 epochs.

Statistical Analysis

Final model performance after training was assessed using AUC, sensitivity, and specificity calculated for the holdout test set. For each model at a given AUC, we reported a sensitive mode (sensitivity level set at 80%) and a specific mode (specificity level set at 80%). The 95% confidence intervals (CIs) for sensitivity and specificity were calculated using the Clopper–Pearson exact method.[17] The 95% CIs for model AUCs were calculated using the percentile method (Python 3: numpy.percentile) after bootstrapping the test set with replacement 10 000 times.[18] To compare the model performance, each model was evaluated on each of the bootstrapped test sets and the resulting AUC scores were compared in a pairwise fashion, resulting in a distribution of AUC difference scores. To identify significant differences in model AUC performance, a modified paired *t* test using the method developed by Nadeau and Bengio[19] was used. This method adjusts the calculated variance of the bootstrapped AUC differences used in the calculation of the T-statistic by a factor of "(1/n + n1/n2)" rather than the traditional "1/n" to account for the nonindependence of the training and test datasets, where n is the total number of CVs or random resamples used (100 in this study), n1 is the number of scans in the test dataset (2051), and n2 is the number of scans in the training dataset (26 622). In this study, we selected a bootstrap random resample number of 100 for n to evaluate significance, rather than the 10 000 previously used to estimate CIs, as proposed by the authors Bouckaert and Frank,[20] to allow for adequate statistical reproducibility without oversampling. We report a range of derived *P* values based on 5 repeat tests and considered $P < 0.05$ to be statistically significant. All model development, testing, and comparisons were performed in the Microsoft Azure ML environment using Python 3.10.8 and PyTorch 1.12.0.

Results

In total, 33 189 OCT volumes from 2084 patients were included in our study. Of these patients, 63% were female.

Table 1. Six Models Were Trained, Tested, and Compared

Model Name	Architecture	OCT	Clinical Variable
MLP multimodal	ResNet-50	Y	Y
CNN OCT	ResNet-50	Y	N
RF multimodal	Random Forest†	Y	Y
RF clinical	Random Forest†	N	Y
XGBoost multimodal	XGBoost‡	Y	Y
XGBoost clinical	XGBoost‡	N	Y

CNN = convolutional neural network; MLP = multilayer perceptron; N = no; RF = Random Forest; Y = yes.
*ResNet-50 is a deep convolutional neural network.
†Random Forest is a classical machine learning model.
‡XGBoost is a classical machine learning model.

The race distribution was 89.1% White, 4.8% Black, 2.3% Asian, 0.1% Native American or Pacific Islander, and 3.8% other. The mean and standard deviation of the age of patients at the time of conversion of the first eye was 78.9 (± 9.3) years.

In total, 6 models (1 from approach 1, 2 from approach 2, and 3 from approach 3) were trained, tested, and compared—MLP multimodal (trained with both OCT and clinical data), CNN OCT (ResNet-50 trained with OCT only), RF multimodal (trained with both OCT and clinical data), RF clinical (trained with clinical data only), XGBoost multimodal (trained with both OCT and clinical data), and XGBoost clinical (trained with clinical data only) (see Table 1). Before integrating information from OCT volumes with clinical data for model training (approach 3), we performed a correlation analysis for these 2 data modalities to assess whether the CNN had identified features from the OCT scans that encompassed the information contained in the clinical data. We found no significant correlations between the feature vector and the clinical data, indicating that the latter could contain additional useful information not found in the OCT volumes.

The performance of all 6 models is summarized in Figure 2A. The models with the best performance were "MLP multimodal" (AUC: 0.76, 95% CI: 0.71–0.80) and "CNN OCT" (AUC: 0.75, 95% CI: 0.70–0.79). A significant difference in AUC performance, with a mean difference of 0.01 ($P < 0.05$), was identified from paired bootstrap resampling of the test dataset, indicating that the "MLP multimodal" model generally had superior performance as compared to the "CNN OCT" model (Table 2). The other 4 models, with classical ML models as backbone, generally had lower performances, with an AUC ranging from 0.70 to 0.72 when both OCT volumes and clinical data were used and an AUC ranging from 0.59 to 0.60 when only clinical data were used. We performed a subgroup analysis to evaluate the performance of our models in predicting imminent first-eye conversions (Fig 2B) and observed a general slight decrease in performance in all 6 models compared to their performance in the prediction of all-eye conversions. For predicting first-eye conversions, the "MLP multimodal" model had a calculated AUC of 0.73 (95% CI: 0.66–0.80) and the "CNN OCT" model had a calculated AUC of 0.71 (95% CI: 0.64–0.78). A significant difference in AUC performance with a mean difference of 0.02 ($P < 0.001$) was identified from paired bootstrap resampling of the test dataset, indicating that the "MLP multimodal" model was superior to the "CNN OCT" model in predicting imminent first-eye conversions as well (Table 2).

Detailed sensitivity and specificity analyses were performed for our top 2 performing models, MLP multimodal and CNN OCT. The performance of these 2 models, when set to a sensitive (80% sensitive) and specific (80% specific) operating point, is summarized in Table 3.

Discussion

In our current study, we compared and contrasted different ways to combine OCT and clinical tabular data into a single predictive ML model. Specifically, we demonstrated in detail how we incorporated clinically relevant biological variables for AMD, such as age and fellow-eye status, into a CNN, which is the most widely used DL architecture for ophthalmic image classification. In general, we observed that CNN-based models outperformed classical ML models,

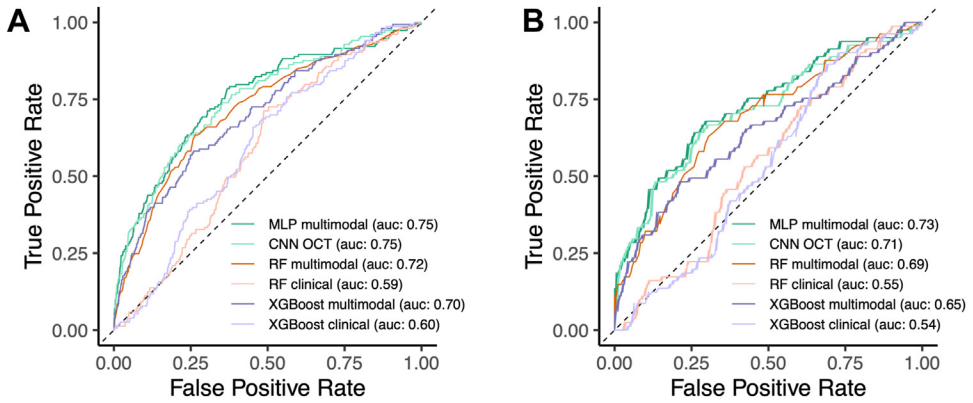


Figure 2. Performance of models. Performance for all 6 models in predicting imminent (within 6 months) conversion to eAMD on an eye level (**A**). Subgroup analysis for predicting first-eye conversions (**B**). CNN = convolutional neural network; eAMD = exudative age-related macular degeneration; RF = Random Forest.

Table 2. Confidence Intervals for Model AUC Scores for All-Eye and First-Eye Conversions

Conversion	CNN OCT	MLP Multimodal	CNN OCT−MLP Multimodal Mean AUC Difference	*P* Values
All-eye conversion	0.749 (0.704−0.792)*	0.755 (0.710−0.797)*	+0.006†	0.008−0.048‡
First-eye conversion	0.714 (0.639−0.781)*	0.730 (0.657−0.795)*	+0.016†	<0.001‡

AUC = area under the operating characteristic curve; CNN = convolutional neural network; MLP = multilayer perceptron.
*95% confidence interval calculated from 10 000 bootstrapped samples of the test set shown in parentheses.
†Significant differences between the "CNN OCT" and "MLP multimodal" models were identified using 5 repetitions of 100 bootstrapped samples of the test set, with AUC differences evaluated by the modified paired *t* test.
‡Statistically significant.

and models incorporating OCT images for training outperformed models that did not incorporate OCT images for training. Not surprisingly, classical ML models trained with clinical tabular data alone performed poorly, with an AUC of approximately 0.6. Our best-performing model, which was DL-based and used both OCT images and clinical tabular data for training (MLP multimodal), showed a reasonably robust performance in predicting imminent conversion to eAMD within 6 months, with an AUC of 0.76. When compared to the "CNN OCT" model, which only used OCT images for training and predictions, the "MLP multimodal" model showed statistically significant superior performance in predicting both all-eye and first-eye conversions. This observation is in line with previous studies that demonstrated that the addition of clinical data to fundus color photographs during model training could improve a ML model's classification performance in the context of chronic kidney disease[21] and AMD detection.[22] During the training of "MLP multimodal," we specifically included the clinical variables of age, sex, BCVA, and fellow-eye status, as these variables were readily available in our electronic health record and have been shown to be biologically relevant variables for AMD.[23–26]

On review of the literature, we found only a handful of studies that utilized DL techniques to predict conversion to eAMD using OCT images. Russakoff et al developed DL models to predict fellow-eye conversion to eAMD over a 2-year period. Although their dataset was small, including only 71 patients, and their model performance was evaluated by fivefold CV instead of the more rigorous way with a holdout test set, they demonstrated that OCT image preprocessing with retinal layer segmentation could further boost model performance.[27]

Banerjee et al[28] used the fellow-eye data from the HARBOR clinical trial (ClinicalTrials.gov identifier: NCT00891735) for model development and took a different approach that utilized a recurrent neural network as backbone and considered multiple time points. Their recurrent neural network used both patient demographic data and quantitative image features as input. The quantitative image features were generated by a proprietary "Advanced RPE Analysis" in the Cirrus Review Software and included 21 drusen specific biomarkers, such as drusen density and drusen volume within the central 3 mm. Their model achieved an AUC of 0.77 in predicting conversion to eAMD within 6 months, a performance comparable to our most robust model. While the inclusion of quantitative drusen biomarkers in their model training improved model explainability, such a priori hand selection of image features also substantially increased the workload for OCT image preprocessing and limited DL's abilities in discovering novel associations that are unknown to human experts.

In the pioneering study by Yim et al,[15] both segmented OCT volumes and whole unsegmented OCT volumes from a large cohort of patients (n = 2795) were used to train a DL model, which was capable of predicting imminent conversion to eAMD within 6 months. Their model achieved an AUC of 0.886 when the date of the first intravitreal injection was defined as the date of conversion and outperformed the majority of a panel of human experts in a head-to-head comparison. However, their model was entirely trained with and tested on fellow eyes of patients with already known eAMD in their first eyes.

While our current study shares some similarities with the aforementioned studies, to our knowledge, our study is the

Table 3. Performance Metrics for the "CNN OCT" Model and "MLP Multimodal" Model Set at 80% Sensitivity and 80% Specificity Operating Points

	CNN OCT		MLP Multimodal	
Conversion	*Sensitivity*	*Specificity*	*Sensitivity*	*Specificity*
All-eye conversion				
Sensitive threshold	79.7% (72.5−85.8)	54.1% (51.9−56.4)	79.7% (72.0−85.8)	60.4% (58.1−62.6)
Specific threshold	57.5% (49.3−65.5)	80.0% (78.1−81.8)	56.9% (48.6−64.8)	80.0% (78.1−81.8)
First-eye conversion				
Sensitive threshold	80.2% (69.9−88.3)	43.7% (36.5−51.1)	80.2% (69.9−88.3)	45.3% (38.0−52.6)
Specific threshold	51.9% (40.5−63.1)	80.0% (73.6−85.4)	53.1% (41.7−64.3)	80.0% (73.6−85.4)

CNN = convolutional neural network; MLP = multilayer perceptron.

first to also test and report specifically a DL model's ability in predicting first-eye conversions, which is valuable from a public health perspective. Patients with eAMD in 1 eye typically have to undergo regular clinic visits for intravitreal anti-VEGF injections, and, therefore, it is already relatively convenient and easy to monitor the status of the fellow eye. As such, while the ability to predict imminent conversion in the fellow eye using DL is interesting from a scientific point of view, it will not drastically alter current practice pattern. In contrast, a model that can predict imminent conversion in the first eye of patients who do not have a prior history of eAMD, if validated prospectively, could change current practice pattern in terms of screening, monitoring, and referral. For example, such a model could be deployed as a clinical decision support tool at primary care clinics to monitor patients with early or intermediate AMD, and these patients will only be referred to a higher level of care if deemed to be at a high risk for imminent conversion. This workflow could in turn allow ophthalmologists and retinal specialists to manage and prioritize patients with more acute ophthalmic issues.

In summary, we developed a methodology to incorporate relevant clinical variables into the training of an OCT-based DL model that is capable of predicting imminent first- and fellow-eye conversion to eAMD within 6 months. We further demonstrated that the incorporation of these clinical variables improved predictions for both all-eye and first-eye conversions. However, our study is limited by the lack of diversity in OCT images, as it only included OCT volumes acquired by the Heidelberg Spectralis platform. Nearly 90% of our dataset was derived from White patients, so it remains uncertain how well our model will generalize to other populations such as Asians, in whom variants of eAMD such as polypoidal choroidal vasculopathy are more common. In terms of clinical variables, we were not able to consider smoking status and genetic variables, as this information were not readily available in our database. Also, our models were not trained to predict the progression from intermediate to late AMD in the form of geographic atrophy—another important feature that causes irreversible vision loss. As the next step, we plan to fine-tune our best-performing model with OCT volumes from other platforms at other institutions using federated learning and compare the performance of our model with human experts in a head-to-head fashion. The SHapley Additive exPlanations test, an approach from game theory used to estimate the influence of a particular feature on the predictive ability of a classifier, could be applied to evaluate which clinical factors improved our models' performance. In addition, we will explore whether the incorporation of known OCT biomarkers for AMD progression, such as central drusen volume, subretinal drusenoid deposits, and intraretinal hyperreflective foci, into model training will further enhance model performance.

Footnotes and Disclosures

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HUMAN SUBJECTS: Human subjects were included in this study. This is a retrospective study that adheres to the tenets of the Declaration of Helsinki and was approved by the Johns Hopkins University Institutional Review Board. Because the study was retrospective, informed consent was not required.

No animal subjects were used in this study.

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Obtained funding: T.Y.A. Liu

Analysis and interpretation: T.Y.A. Liu, Y. Liu, Gastonguay, Midgett, Kuo, Zhao, Ullah, Alexander, Hartman, Koseoglu, Jones

Overall responsibility: T.Y.A. Liu, Y. Liu, Gastonguay, Midgett, Kuo, Zhao, Ullah, Alexander, Hartman, Koseoglu, Jones

Abbreviations and Acronyms:
3D = 3-dimensional; **AMD** = age-related macular degeneration; **AUC** = area under the operating characteristic curve; **BCVA** = best-corrected visual acuity; **CI** = confidence interval; **CNN** = convolutional neural network; **CV** = cross-validation; **DL** = deep learning; **eAMD** = exudative age-related macular degeneration; **ML** = machine learning; **MLP** = multilayer perceptron; **RF** = Random Forest; **VA** = visual acuity.

Keywords:
Age-related macular degeneration, Deep learning, Ensemble machine learning, Multimodal data, Optical coherence tomography.

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